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An optimized anomaly detection framework in industrial control systems through grey wolf optimizer and autoencoder integration

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Ensuring reliable Internet connectivity in Industrial Control Systems is critical for real-time monitoring and anomaly detection. Existing methods, however, struggle with high computational complexity, limited applicability to specific datasets, and elevated false-positive rates. This paper presents a novel collaborative data processing framework that enhances anomaly detection in ICS by integrating the Grey Wolf Optimizer with Autoencoders. The proposed approach optimizes GWO by improving prey selection, encircling mechanisms, and initial population generation, while enhancing AE dropout functionality for improved model generalization. The method operates in two stages: (1) Optimizing GWO for feature selection to identify relevant features and reduce feature errors, and (2) Utilizing AE for efficient anomaly detection. Experimental validation on the SWaT and WADI benchmark datasets demonstrates the superior performance of the proposed model, achieving significant improvements in accuracy, precision, recall, and F1-score over existing state-of-the-art approaches. These results highlight the potential of the proposed approach in addressing the limitations of current anomaly detection systems in ICS.

Keywords Industrial control system, Security, Anomaly detection, Collaborative approach (GWO+AE), SWaT, WADI datasets

The advancement of technology and the industry's transformation from manual to automated and Industry 4.0 are exposing several vulnerabilities in Industrial Control Systems (ICS). These vulnerabilities serve as gateways for attackers to compromise the industry or process, raising concerns about ICS security through cyber attacks that can cause financial or physical damage. Furthermore, the study reveals that the number of cyberattack incidents in ICS is increasing rapidly, as seen in the 2010 Stuxnet attack, the 2015 Ukraine power grid attack, and many others studied in[1]. According to the ICS-cyber emergency report[2], the number of cyber attacks is increasing significantly.

These serious incidents show that ICS security is challenging and necessary. This requirement of ICS security, based on user requirements, is a means to access the powerful methodologies of the ICS environment. Machine Learning (ML) and Deep Learning (DL) are important in various domains, such as fraud detection, image recognition, and anomaly detection in ICS. Researchers have used various ML/DL approaches to secure the ICS environment.

Recently, researchers have proposed various anomaly detection approaches in[1,3–5]. For instance, the authors in[6,7] used a deep learning-based approach for anomaly detection in ICS using the SWaT dataset. Still, the work suffers from a high false positive rate and computational complexity. In another work, the authors used deep learning approaches for anomaly detection in ICS using the BATDAL dataset. Still, outcomes show that work suffers from a high false positive rate, low accuracy, and computational complexity. Similarly, various research works use ML approaches for anomaly detection in ICS, but still, there is a lack of a generic model for anomaly detection.

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Recently, the researcher was drawn to the DL approaches for anomaly detection in ICS. DL is a useful and more powerful ML approach that is crucial in identifying the most correlated features from large datasets. Various models have been proposed in the literature, including Recurrent Neural Networks, Denoising Autoencoders (DAE)[8–10], Deep Belief Networks, Convolutional Neural Networks (CNN), and Deep Neural Networks (DNN). Here, outcomes show that AE-based approaches are effective for securing the ICS environment[7], while other approaches, such as CNN and Long Short-Term Memory (LSTM), are more suitable for image-based datasets.

This paper proposes an approach for anomaly detection in ICS. In this work, a collaborative approach combining unsupervised learning with an Autoencoder (AE) and a metaheuristic approach, Grey Wolf Optimizer (GWO), is employed.

Motivation and contribution

Motivated by the above researcher's contribution, various works have been done on anomaly detection in ICS using DAE, LSTM, CNN, AE, and RNN approaches. However, these approaches are insufficient for anomaly detection in time series datasets due to their low accuracy and high false positive rates. Due to the complex nature of the ICS environment, available approaches are insufficient to overcome the security challenges. To overcome such challenges, an efficient anomaly detection approach is required that carefully examines both real-time and historical data streams with sufficient accuracy, a low false positive rate, and minimal computational complexity. However, the anomaly detection approach specifically for ICS environment data is studied in this work. Important in this work to select highly correlated features from the available datasets of SWaT and WADI and classify them into anomalous classes. For feature engineering, we used the GWO approach, which is based on evolutionary computation, and a detailed study is included in the section below. This approach has been used due to its flexibility, simplicity, and optimization power, which authors recently utilized to optimize IoT network connectivity[11]. We used the AE approach for anomaly detection, a promising DL-based approach. Furthermore, the approach used enhances the studied model by improving the capabilities of exploration, exploitation, and initial population generation for GWO, while also increasing the AE functionality by removing irrelevant features. The enhanced AE drop-out functionality helps avoid overfitting and enhances the weights of relevant features, which enhances accuracy.

The researcher's work has shown that AE and GWO are valuable and powerful approaches. In[11], the authors used GWO to optimize IoT network connectivity in WSNs. In this experiment, first, this work used GWO for anomaly detection in ICS using the SWaT dataset, which assists in yielding sufficient accuracy, and then collaborated our GWO and AE model; previously in this work, and used AE for anomaly detection in ICS, which also resulted in yielding sufficient accuracy and reduced the high false positive rate. In this work, we collaborated with GWO and AE, adopting a more robust and valuable approach that enhances accuracy, reduces computational complexity, and lowers the high false positive rate.

Here, to highlight the key contributions of the proposed work, which are detailed below in a few steps, along with significant achievements.

- This work developed a new, useful collaborative approach for anomaly detection in ICS. The collaborative model was developed using the GWO and AE approaches. GWO is used for feature engineering and selecting highly correlated features, and AE is used for anomaly classification.
- In this proposed GWO approach, it is used to optimize the exploration, exploitation, and initial population generation. In optimised GWO, outcomes are improved over those in basic GWO.
- SWaT and WADI benchmark datasets have been used to prove the proposed work. The proposed model yielded sufficient outputs with improved accuracy of up to 0.9996 for SWaT and 0.9975 for WADI, reduced high false positive rates to 0.0005 for SWaT and 0.0007 for WADI, and achieved precision of 0.9996 for SWaT and 0.9919 for WADI.
- There is a comparison of results with state-of-the-art approaches on SWaT and WADI datasets, which shows that the proposed model is improved and yields sufficient results.

The remainder of this paper is organized as follows: "Related work" presents the related work on anomaly detection in ICS. "Proposed methodology" introduces the proposed methodology. "Materials and methodology" describes the materials and methods, including data preprocessing, feature engineering, and model training. "Simulation and results" reports the simulation setup, results, and analysis. Finally, "Conclusion and future work" concludes the paper and outlines potential directions for future research.

Related work

ICSs were traditionally used in solitary settings, with an emphasis on safety, and each system was designed to halt the process in the event of an error[12]. However, there is now significantly less separation from the outside world, thanks to the emergence of Internet protocols, IoT devices, and wireless technology within ICSs. Because safety systems were not intended to withstand malevolent attacks, they are more vulnerable than before.

Authors in[13] studied the concept of the first Intrusion Detection System (IDS). The authors in this work explained the solution for IDS. It employs a dual strategy that uses a rule-based expert system to identify understood intrusion types. It also utilizes a statistical anomaly detection component that leverages target systems, user profiles, and host systems. The same research team subsequently developed an upgraded version called the next-generation intrusion detection expert system[14]. Following the release of the MIT and DARPA Intrusion Detection Evaluation[15] in 1998 and 1999, the concept of using anomaly detection for information security gained attention. However, as[16] demonstrated, DARPA datasets are not suitable for modeling real-world network systems. As a result, creating new datasets is essential for IDS development.

The methods for statistically detecting anomalies identify items in the low-probability zone of the model as outliers after learning a generative model that matches a given dataset. In[17], the authors extract their Statistically Smoothed Trajectory (SST) features in time series data. They then model the appropriate parameters using a set of nonlinear functions to find anomalies. Authors have proposed a statistical approach for anomaly detection in[18]. The proposed approach is based on Tukey and relative entropy statistics for data monitoring from previous periods. To identify discordances of different lengths in a time series, the authors in[19] presented the MERLIN anomaly detection algorithm, which combines distances to analyze each subsequence as it slides along the sequence window.

Various ML-based approaches have been used for anomaly detection in ICS. Based on isolation forest (iForest), the authors of[20] provided a sequence data anomaly detection technique. A random feature selection process constructed a binary tree for each sample. Near the root, the data was determined as abnormal, whereas farther away from the root, it was normal. The authors in[21] presented a distance-based, unsupervised outlier detection approach that uses the total of a point's k closest neighbors' distances to identify outliers. An HTM-based approach for multivariate time series anomaly detection has been examined in[22]. This method handles data in multivariate time series by employing multiple spatial pools, rather than the conventional structure with a single spatial pool and encoder. In[23], the scientists built a behavioral model to identify semantic attacks that relied on secondary labeling of the states-duration evolution graph and the state transition probability graph.

With the development of technology, authors have used various deep-learning approaches for anomaly detection in ICS. An AE-based attack detection approach has been employed by the authors in[24] to extract the original unbalanced dataset, creating a new representation that is then optimized for AE, DNN, and DT, respectively. It employs Deep Neural Networks (DNNs)[25] and Decision Trees (DTs) to identify network attacks. The output of a position in the previous stage may not be suitable for tasks in the subsequent stage because these approaches lack effective communication and feedback across subtasks. To avoid performing feature engineering and classification optimization separately, the authors of[26] presented a novel optimization technique that enables the simultaneous optimization of multiple tasks. The authors developed joint DR and K-means clustering approaches for suitable cluster data, utilizing DR with DNN. In another work[27], authors used a dual AE-GAN-based anomaly detection approach for anomaly detection in ICS. With the same harmful attack effect, this tactic can increase the likelihood that adversarial samples will evade the detector. An attacker's goal in the ICS context is to conceal cyber-physical attacks from IDS detection by employing adversarial attack techniques. For experimental work, the authors used the SWaT dataset. Authors in[28] used GRU-Based Interpretable Multivariate for anomaly detection in ICS. For the experimental setup, the authors used SWaT and WADI datasets. Using the SWaT dataset, the authors obtained an accuracy of 0.8845, with a recall of 0.5909 and an F1-score of 0.7494. The authors achieved results with a precision of 0.9968. Meanwhile, the WADI dataset attained an accuracy of 0.7821, with a precision of 0.3584. These outcomes show that the model is incomplete for anomaly detection in ICS. Researchers in[24] used a DL-based approach for anomaly detection in ICS. The authors used DNN and Decision Tree (DT) classifiers for anomaly detection in ICS. The authors named this approach a generic model for a less computationally complex model. For experimental work, gas pipeline and SWaT datasets have been used. The SWaT dataset's accuracy is 0.9867, with a precision of 0.785, a recall of 0.624, and an F1-score of 0.887. On the other hand, with the gas pipeline dataset, the outcomes are reported as 0.856 and an F1-score of 0.617. It shows that the model can not be generic for anomaly detection in ICS. Still, there are high false positive rates, which need to be reduced. The authors in[29] used a deep learning-based approach for network IDS. A non-symmetric deep autoencoder (NDAE) approach has been used for feature learning. For experimental work, the authors used KDD Cup '99 and NSL-KDD datasets. Using KDD CUP'99, the obtained accuracy was 97.86% , with a recall of 97.75% and an F1-score of 97.15% . The accuracy using the KDD CUP'99 dataset was 85.42% , with a recall of 80.54% and an F1-score of 84.08% . The work yielded promising results, but it cannot be a generic model, and accuracy needs to be enhanced with other metrics. In another work, the authors propose a genetic algorithm to detect neural network architectures using the provided dataset, employing the NAB metric to assess the quality and consistency of various architectures. The authors used the SWaT dataset for the experimental setup. It resulted in an F1-score of 0.812 and a Recall of 0.702. Researchers in[30] used an unsupervised ML-based approach for anomaly detection in ICS. A one-class SVM approach has been implemented on the SWaT dataset. The work has been compared with the DNN-based model, demonstrating that the DNN-based model performs slightly better than the SVM-based model. In terms of outcomes, it has been shown that the SVM approach is somewhat more complicated for anomaly detection in ICS than the DNN approach. The authors used the SWaT dataset. Further detail and comparison of the related work is given in Table 1.

Several recent studies have investigated hybrid two-tier frameworks that integrate metaheuristics with machine learning or deep learning to enhance prediction accuracy. The authors in[31] proposed a dual-layer deep learning and machine learning approach enhanced by an adaptive multi-population firefly method for forecasting software errors. Two-tier metaheuristic frameworks have been employed for tasks such as natural language processing[32], intrusion detection in IoT systems facilitating the metaverse[33], and phishing detection, which combines feature selection and XGBoost optimization[34]. Moreover, metaheuristic-optimized XGBoost models have demonstrated efficacy in forecasting software defects, using SHAP values to elucidate the outcomes[35]. These tests demonstrate that multi-layered, optimization-augmented learning frameworks are effective. We employ this technology and modify it to detect anomalies in Industrial Control Systems.

However, this section highlighted some works for anomaly detection in the ICS environment. The majority of the studied works are based on feature engineering. Studied approaches are computationally complex and less accurate, with a high false-positive rate. There is a need for a novel and generic approach to reduce the high false positive rate, and it should be less computationally complex and more accurate for anomaly detection in ICS. Moreover, most currently proposed approaches are tested on balanced datasets, which do not accurately

Study	Detection method	Learning type	Dataset used	Key contributions	Main limitations
[36]	Deep autoencoder (DAICS)	Supervised	SWaT	Deep neural pipeline for anomaly detection in ICS	No feature selection or interpretability support
[24]	Ensemble deep learning	Supervised	Custom ICS	Improves detection using model ensemble	Dataset-specific, lacks generalization
[37]	Multi-modal explainable DL	Supervised	Custom ICS	Explainable model with multimodal data fusion	High model complexity and deployment overhead
[27]	Dual AE + GAN	Unsupervised	SWaT	Detects unknown attacks via GAN-enhanced AE	High computational cost for training
[38]	GNN-based anomaly detection	Unsupervised	SMD, SWaT	Captures graph structure across sensors	Requires large labeled datasets and long training time
[7]	Improved AE model	Semi-supervised	SWaT, WADI	Better feature reconstruction and anomaly scores	No adaptive feature optimization
[39]	Federated DL	Supervised	Simulated CPS	Secure collaborative learning with scalability	High communication and synchronization overhead
[38]	Adversarial SVDD + GNN	Supervised + Adversarial	ICS-ST, SMD	Robust adversarial anomaly defense using graph structures	Complex and computationally expensive
[40]	Dual-variational GAN	Semi-supervised	CPS testbeds	Generalizes well to novel attack scenarios	Poor interpretability of GAN models
Proposed work	GWO-based feature selection + AE with explainability	Semi-supervised	SWaT, WADI	Integrates optimization and interpretable anomaly detection, scalable across ICS domains	Not yet evaluated on cross-domain or federated setups (planned future work)

Table 1. Harmonized comparison of recent ICS anomaly detection studies and proposed framework.

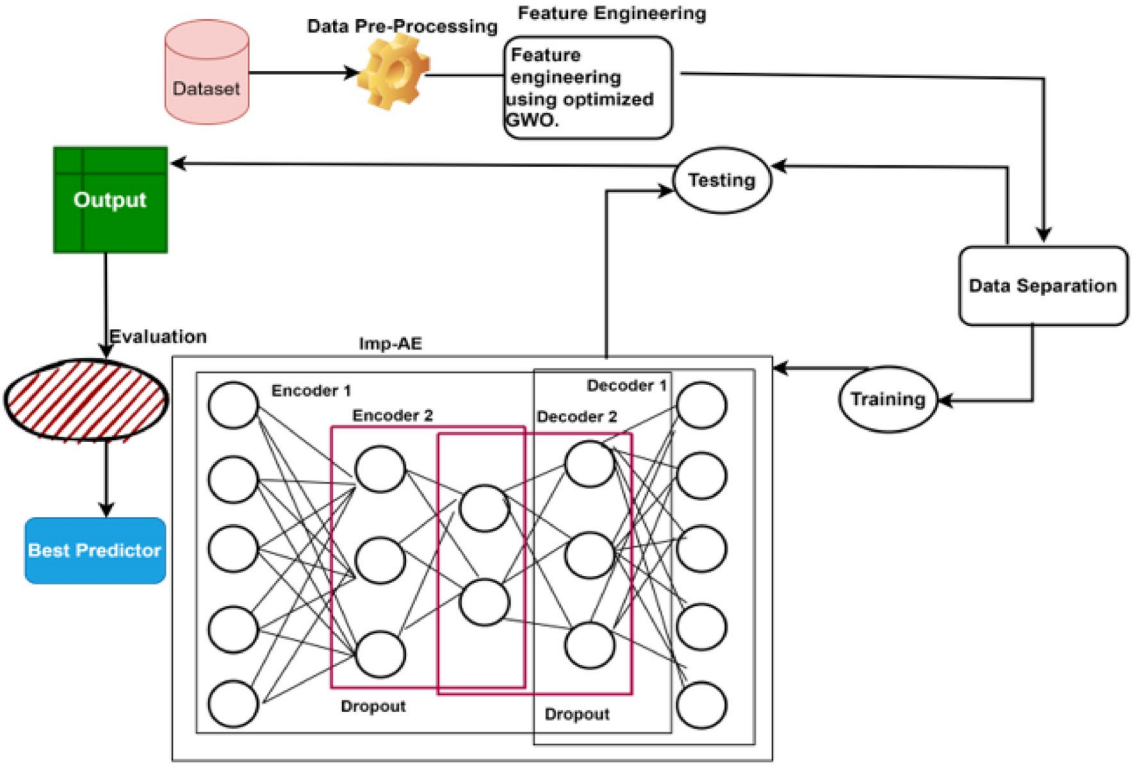


Fig. 1. Proposed model.

represent the typical ICS environment, where data is often imbalanced. Their deployment is challenging because these algorithms cannot extract different discriminative information from real-world imbalanced datasets.

Proposed methodology

This section highlights the proposed approach for anomaly detection in ICS. The following steps outline the data processing procedure in the experimental setup. First of all, an important thing is the selection of a dataset for experimental setup; the second step is pre-processing of the available dataset, and the third step is feature engineering with optimized GWO and selection of high co-related features followed by data separation splitting into testing and training, model selection, and output, and the last step is evaluation, anomaly detection, and best prediction using AE approach as shown in Fig. 1. Now is the time to explain the experimental setup.

The first step of the studied collaborative approach is dataset selection. Here, we used the SWaT dataset. From this dataset, finding anomalies or unusual patterns in the system's behavior can be simplified by a collaborative approach. The second step is data pre-processing, which involves handling categorical variables, addressing

missing values, and normalizing the data. The next step is feature engineering, where highly correlated features are extracted based on the generated heatmap of the available dataset, due to its hunting behavior. GWO helps find highly co-related features, which uses the GWO approach and generates a heatmap and highly co-related features, which assists in developing an efficient model, then is data separation into 20% training and 80% testing, the training data will be helpful for anomaly detection in a collaborative model, and testing is for performance evaluation. Next in model selection is the VAE model, which learns to reconstruct its input data and is valid for anomaly detection. Here, trained a VAE-based model to reconstruct regular data instances and identify anomalies. Next, the trained AE model is applied to the testing dataset to yield predictions for each instance. High reconstruction errors or deviations in the system will be considered as anomalies, and next, the proposed model's performance will be evaluated using precision, recall, F1-score, and accuracy metrics. The feature extraction capability of GWO, combined with the anomaly detection of the VAE model, makes it a collaborative model. This collaborative approach is efficient for anomaly detection.

Selection and optimization of algorithm parameters

The GWO algorithm is crucial for feature selection and enhancing the efficacy of anomaly detection in this proposed method. Grounded in theoretical principles and empirical testing, we selected the GWO parameters, including population size, maximum iterations, and specific configurations for prey selection, encircling processes, and initial population generation. Multiple factors influenced the use of GWO for feature selection in this study. In this work, the authors employed the GWO to identify the most significant features from the input dataset. The objective function was the F1-score derived from a validation set utilising an AE trained on the selected features. The algorithm was executed 50 times, with a population size of 30 grey wolves established. We executed the GWO independently ten times to ensure its robustness, and thereafter, we selected the feature subset that exhibited the optimal average validation performance. In the solution encoding technique, each individual (grey wolf) was represented as a binary string, with a length corresponding to the number of features, where '1' indicated that the feature was selected and '0' meant it was not. Inspired by the social nature of grey wolves, the GWO is a robust evolutionary algorithm characterized by its superior global search capabilities and fewer tuning parameters than other evolutionary algorithms, such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO). GWO is highly effective for addressing complex optimization problems, especially those involving high-dimensional data sets, such as feature selection in industrial control systems, due to its equilibrium between exploration and exploitation. Due to their use in optimization tasks and their efficacy in addressing non-linear, high-dimensional problems, we examined GA and PSO as potential alternatives. However, for the subsequent reasons, we prefer GWO over GA and PSO.

- GA Vs GWO: Despite being a recognized optimization technique, GA frequently necessitates more advanced operators, such as crossover and mutation, and more refinement to achieve optimal outcomes. Consequently, its reliance on evolutionary processes may result in a slower convergence rate for GA in certain circumstances. In contrast, GWO is a simpler and more efficient approach for feature selection in this context[41]. GA uses natural selection mechanisms to evolve solutions. The fitness function for GA can be represented as:

$$f(x) = \sum_{i=1}^{n} w_i \cdot x_i \tag{1}$$

Where: w_i are the weights of the features, x_i are the feature values. GA relies on three leading operators—selection, crossover, and mutation—to evolve feature subsets.

- PSO Vs GWO: In continuous optimization problems, PSO is distinguished for its swift convergence. The particle-based search methodology can occasionally result in PSO becoming ensnared in local optima[42,43]. GWO is more suitable for the feature selection process in ICS anomaly detection as it circumvents this issue and offers a more balanced exploration-exploitation strategy through its predation-inspired search method[44,45]. PSO is inspired by the movement of particles in a search space. The position update rule for PSO is:

$$v_i^{new} = w \cdot v_i + c_1 \cdot r_1 \cdot (p_i - x_i) + c_2 \cdot r_2 \cdot (g_i - x_i) \tag{2}$$

$$x_i^{new} = x_i + v_i \tag{3}$$

Where: v_i is the velocity of the ith particle, x_i is the position of the ith particle, p_i is the best position of the ith particle, g_i is the best position of the swarm, c_1, c_2 are acceleration constants, and r_1, r_2 are random values.

Performance evaluation

For performance comparison, the following metrics were used:

- Accuracy: The proportion of correct predictions over all predictions.
- Computational time: The total time taken by each algorithm to converge and optimize the feature selection process.
- Convergence rate: The number of iterations or generations required to reach an optimal or near-optimal solution.

Grey wolf optimizer (GWO)

This section highlights the basic GWO model. GWO is a meta-heuristic approach that is inspired by wolf behavior. GWO belongs to the Canidae family. GWO is considered the top of the food chain, and they prefer to live in groups or packs. The average group size is 5-12 wolves in a pack. Algorithm 1 provides the Optimization of GWO in proposed approach.

α are considered leaders in the pack and can be male or female. It can decide on prey, sleeping time, waking time, and place, among other things. α is known as dominant in the group, and all wolves have to follow his/her instructions. Because of some democratic behavior in the pack, α doesn't need to be the strongest, but it is considered best in terms of management, which shows the strength of discipline and organization in the pack. The second-level wolves are known as β, and the subordinate group assists the α in decision-making. If one of the α is in emergency or unavailable, the β takes their place. The β wolves follow the α but have command over the lower-level wolves. Its role can be considered an advisor or spokesperson of α. It shares the feedback with α. The third is γ, which follows the instructions from α and β and relies on the δ. The δ are the lowest level wolves that follow the above α, β, and γ.

The social hierarchy of wolves involves tracking, chasing, and approaching prey, then pursuing, encircling, and harassing it until it stops moving, and finally attacking it. This social hierarchy mathematically can be represented as

Require: Dataset T (maximum number of iterations), *pop* (population size), F (number of features), *pos* (grey wolf position)
Ensure: Output: Highly correlated features
Initialize parameters
Generate initial population based on function distribution
Initialize grey wolf position $\vec{A}$
Compute r_2 from Eq. 7
Initialize $\vec{C}$ using r_2
Compute fitness function using Eq. 9
Set α as the grey wolf with the best fitness
Set β as the second-best grey wolf with fitness
Set δ as the third-best grey wolf
while $t < T$ **do**
  for $i = 1$ to *pop* **do** ▷ For each grey wolf in the population
    for $j = 1$ to F **do** ▷ For each feature
      Compute p_m using Eq. 19
      if $p_m > r_1$ **then** ▷ Non-linear function check
        Compute $\mathcal{Z}$ using Eq. 19
        Update the fitness for features: $\{f_1, f_2, f_3, \ldots, f_N\}$
        for $K = 1$ to N **do**
          Reinitialize the k-th feature based on the new fitness
        end for
      end if
    end for
  end for
end while
Return the most correlated features after optimization

Algorithm 1. Optimized GWO for feature engineering.

Social hierarchy

When designing GWO, the social hierarchy of the GWO is considered in the mathematical model, where α is considered the best solution. Similarly, β and γ are second and third-best solutions, respectively. The rest is δ. While α, β and γ inspire for hunting. The δ follows all three other solutions.

Encircling

Mathematically, the encircling behavior of wolves is given as

$$\vec{D} = \vec{C} \cdot \vec{X}_p(t) - \vec{X}(t) \tag{4}$$

$$\vec{X}(t+1) = \vec{X}_P(t) - \vec{A} \cdot \vec{D} \tag{5}$$

Here, $\vec{X}$ represents the grey wolf position vector, $\vec{X}_P(t)$ is the prey position vector, t is for iteration, and $\vec{A}$ and $\vec{C}$ are the coefficient vector. We can calculate $\vec{A}$ and $\vec{C}$ as:

$$\vec{A} = 2\vec{a} \cdot \vec{r_1} - \vec{a} \tag{6}$$

$$\vec{C} = 2.\vec{r_2} \tag{7}$$

Here, $\vec{r_1}$ and $\vec{r_2}$ are random vectors [0,1] and $\vec{a}$ components are linearly decreased from 2 to 0. Here, the effects of the Eqs. 4 and 5 are inspiring and are two-dimensional position vectors. It shows that a wolf can vary according to the prey's position. Various points in suitable agents may access the current position by adjusting $\vec{A}$ and $\vec{C}$ values. The variation can be accessed by $\vec{A}$= 1,0 and $\vec{C}$=1,1 while r_1 and r_2 permits wolves to access any position. Grey wolves can be in the same position according to Eqs. 4 and 5.

Hunting
Grey wolves can track down and surround their prey using their keen sense of location. Usually, the alpha leads the hunt. On rare occasions, the beta and delta may also take part in hunting. Nevertheless, we are unsure of the optimal (prey) location in an abstract search space. To imitate the hunting behavior of grey wolves numerically, we assume that the alpha, beta, and delta have superior knowledge regarding the possible location of prey. To force the other search agents, including the omegas, to adjust their positions to the positions of the top search agents, we preserve the top three results thus far. Mathematically, we can represent it as

$$\vec{D}_\alpha = |\vec{C}_\alpha \cdot \vec{X}_\alpha - \vec{X}|, \vec{D}_\beta = |\vec{C}_2 \cdot \vec{X}_\beta - \vec{x}|, \vec{D}_\delta = \vec{C}_3 \cdot \vec{X}_\delta - \vec{X}| \tag{8}$$

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \cdot (\vec{D}_\alpha), \vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \cdot (\vec{D}_\alpha), \vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \cdot (\vec{D}_\delta) \tag{9}$$

$$X(\vec{t+1}) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \tag{10}$$

It shows that prey will be in random places and circle by $\alpha, \beta, and \delta$. It can be said that $\alpha, \beta, and \delta$ guess the prey position, and the remaining wolves surround the prey in a circle.

Exploitation
Exploitation in GWO is used to attack prey. After prey, wolves stop moving. mathematically we reduced $\vec{a}$. Value of $\vec{A}$ (a random value [-2a,2a])is reduced by $\vec{a}$. Here, the value of a reduces from 2 to 0. In the case of $\vec{A}$ -1,1, the next wolf position can be at any point or position between the current and prey position.

Exploration
Grey wolves follow the dominant one, searching for the position of $\alpha, \beta, and \delta$. They separate to find prey and then come together to attack it. $\vec{A}$ will be greater than one or less than -1. $|A| > 1$ pushes the wolves to diverge from prey to better prey. Another GWO agent suits the exploration $\vec{C}$ as in Eq. (7) consists of a random value [0,2] that provides a random weight for prey $C > 1$ or $C < 1$, which is called emphasizes or de-emphasizes, respectively, the prey effect as in Eq. (4). The main objective of the GWO is to resolve the optimization challenge.

- The proposed approach helps the GWO obtain an efficient solution over the iterative process.
- Encircling represents the nearby circle around the solution to extend a higher dimension as a hypersphere.
- It helps the candidate find a suitable position.
- a and A are suitable for adopting exploitation and exploration.
- GWO can smoothly switch between exploration and exploitation thanks to the adjustable values of parameters a and A.
- a and C are the only parameters for the adjustment of GWO.

Combining mutation with other evolutionary processes may make it possible to replicate the entire life cycle of grey wolves. To minimize the number of operators that need to be changed, we have simplified the GWO algorithm. There is a suggestion of future research on such mechanisms.

Anomaly detection strategy in ICS

Security in ICS is a key challenge. For instance, Crashoverride physically damages facilities by deploying fraudulent control procedures. As such, a payload anomaly detection technique is essential. The payload data of an ICS network packet is continuous and correlated with that of other packets. As a result, by using a deep learning technique to detect anomalous behavior, it can anticipate the payload with a high degree of precision. Machine learning and deep learning approaches are essential in securing the ICS environment. DL approaches require more data for the training process, consistently presume a stable environment, and cannot discern between correlation and causation. Despite these challenges, DL is a valid approach for anomaly detection in ICS. Training data can be gathered when there is a higher level of communication in an ICS environment. In the absence of an attack, ICS communication becomes stable, and the transmitted data is related. However, deep learning approaches can be helpful for anomaly detection in ICS.

It is challenging to gather the attack traffic data in an ICS environment compared to normal traffic data. Some classical approaches, such as CNN and RNN, are not suitable for anomaly detection in ICS because they struggle to detect the correlation between measured data. Because of this, AE is considered ideal for anomaly detection in

ICS because of its data correlation. Additionally, AE can train models using unsupervised learning and operate on low-power machines. An autoencoder can perform nonlinear dimension reduction, which is more effective at identifying anomalies than other dimension reduction techniques, such as PCA, which only employ linear dimension reduction.

Require: Dataset x_t, y_t (input data for training), Z_t (autoencoder latent space), V_{thre} (threshold for anomaly detection)
Ensure: Anomaly detection results
Initialize Autoencoder architecture
Train the Autoencoder using dataset x_t and y_t
Compute reconstruction error $e = |x_t - \hat{x}_t|$ where $\hat{x}_t$ is the reconstructed output
Set anomaly threshold V_{thre} based on training data
for each test sample **do**
    Compute reconstruction error e
    if $e > V_{\text{thre}}$ **then**
        Mark the sample as an anomaly
    else
        Mark the sample as normal
    end if
end for
: Return the anomaly detection results

Algorithm 2. AE-based anomaly detection.

Materials and methodology

Details of the used dataset and studied approach for anomaly detection in ICS for this work are given below:

Dataset

This section introduces the dataset used. Secure Water Testbed (SWaT), a fully functional water treatment plant, consists of 51 tags divided into 25 sensors and 26 actuators. The SCADA workstation monitors the water treatment process and is divided into six small processes. These processes are Raw water supply and storage, ultrafiltration, and backwash. Here, PLCs are used to control these sub-processes. Sensors are used for communication (Level indicator, water flow indication transmission, analyzer indicator, pH analysis), and actuators like motor valves and pumps. The PLCs provide actuation orders to the actuators based on the data they receive from the sensors that track the physical process. Such commands can be used to maintain or modify an actuator's present state, depending on the underlying logic of the PLC. At the author's request, these datasets are provided by the iTrust Center, Singapore University of Science and Technology, Singapore.

The dataset comprises 946,722 recordings of sensors and actuators collected at a rate of one sample per second over 11 days of operation. There are 51 properties in each record, which correspond to 25 sensor readings and 26 actuator states. The dataset consists of two parts: the training set, which comprises 7 days of regular operations, and the test set, which includes 4 days of regular activities plus 36 attacks generated by applying the attack model in[7]. Validation is performed on 20% of the training set (validation set). The test set comprises attacks that can affect one or more devices simultaneously and with durations varying from two minutes to nine hours. Spoofing sensor data or switching actuator operation modes are examples of attacks. For example, in the first attack, a motorized valve is set to open for 15 minutes after it has been closed, allowing the tank to overflow. Another approach aims to overflow a tank by decreasing the value recorded by a water level sensor by 0.5 mm/sec for seven minutes. Another dataset used is named the WADI dataset. An actual water distribution network is scaled down in the WADI testbed. The testbed network is divided into three sub-processes, or stages: a return-water grid that manages the extra water from the consumer tanks, a secondary grid that distributes water to six consumer tanks, and a primary grid for water delivery. Just like the SWaT testbed, a SCADA workstation also oversees WADI. Additionally, a PLC that interfaces with actuators and sensors controls each subprocess. Water flow indication transmitters, level indicator transmitters, analyzer indicator transmitters, and pressure meters are examples of sensors. Actuators include motorized valves and pumps, similar to those found in the SWaT testbed. Sixty-nine sensor readings and 54 actuator states, totaling 123 characteristics, were collected over 16 days to form the 1,209,610 records that comprise the WADI dataset. Within the first 14 days, normal operating conditions were recorded in two sets: 95% for training and 5% for validation. The test set spans the last two days and includes 15 attacks and usual activity. Despite having a shorter duration (ranging from two to thirty minutes), the attacks adhere to the same model as those found in the SWaT dataset and are detailed in[7,46]. One attack targets the quantity of chemicals in the water by shutting off a sensor and transmitting false readings to the PLC for 10 minutes. Another attack cuts off the water supply to customer tanks.

This work presents a comprehensive methodology that encompasses all critical phases of its experimental technique, including data preprocessing, addressing class imbalance, data partitioning, and outcome evaluation.

Stage	Details
1. Data cleaning and Preparation	Missing or invalid sensor readings were removed or interpolated
2. Feature normalization	Continuous features normalized using min–max scaling Discrete/binary features one-hot encoded (if applicable)
3. Handling imbalance	Applied SMOTE (synthetic minority over-sampling technique) to oversample attack samples Class weighting used during model training to balance loss function
4. Feature selection	GWO-based feature selection used to reduce irrelevant/noisy features before model training
5. Data splitting	20% training and 80% testing
6. Model training and tuning	Models trained using the training set Hyperparameters tuned using validation set (grid/random search)
7. Evaluation metrics	Evaluated on test set using: precision, recall, F1-score Each experiment repeated 5 times; mean and standard deviation reported

Table 2. Experimental protocol: preprocessing, data splitting, and evaluation workflow.

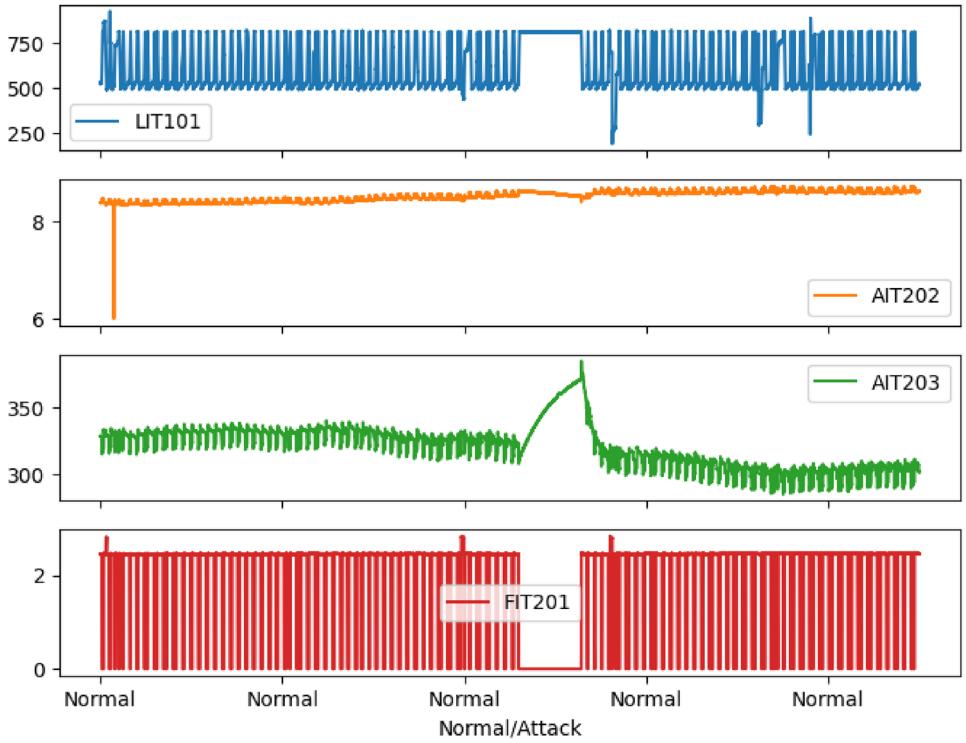


Fig. 2. Observation from SWaT features.

This ensures that this work is reproducible and comprehensible. Table 2 provides a concise summary of these procedures. For all investigations, including the SWaT and WADI datasets, this structured pipeline was consistently employed. It addresses acknowledged issues, such as label imbalance and discrepancies in feature scales, while ensuring that model evaluation is equitable and robust.

Feature engineering

Classifier performance depends heavily on the selection of features. This problem consists of highly correlated features to maximize performance. Suppose, $D = c_1, c_2, c_3, \ldots\ldots c_n$ be a dataset is available dataset with objects n and feature set $F = a_1, a_1, a_3, \ldots\ldots, a_m$, where m represents the features. The mapping of $W(D, F, A_{de}) \rightarrow F'$. $F\cdot$ is considered a feature selection algorithm, and A_{de} is known as class labels, which can be called a decisive attribute. Selected feature are $F' \subset F$ while $|F' = || = k(k < m)$. The primary objective of feature engineering is to refine and select highly correlated features in the dataset, as illustrated in Figures 3 and 4, which represent the graphical representations of the SWaT and WADI datasets, respectively, along with their feature correlations. Correlation is indicated through color in the heatmap, where blue (cooler colors) represent negative correlation, dark colors represent stronger correlation, and red (warmer colors) represent positive correlation between features. The color intensity represents the magnitude of the correlation. PIT503 is for reverse osmosis and shows the graphical representation of the pressure gauge in this data. The PIT503 sensor exhibits the highest correlation. PIT501 and FIT201 are in strong relation. Figure 2 is a graphical representation of SWaT's highly correlated features in correlation. It makes sense that the PIT501 sensor is to dechlorinate the water to P5. In an ICS environment, such correlation is very complicated when each sensor and actuator works independently. A

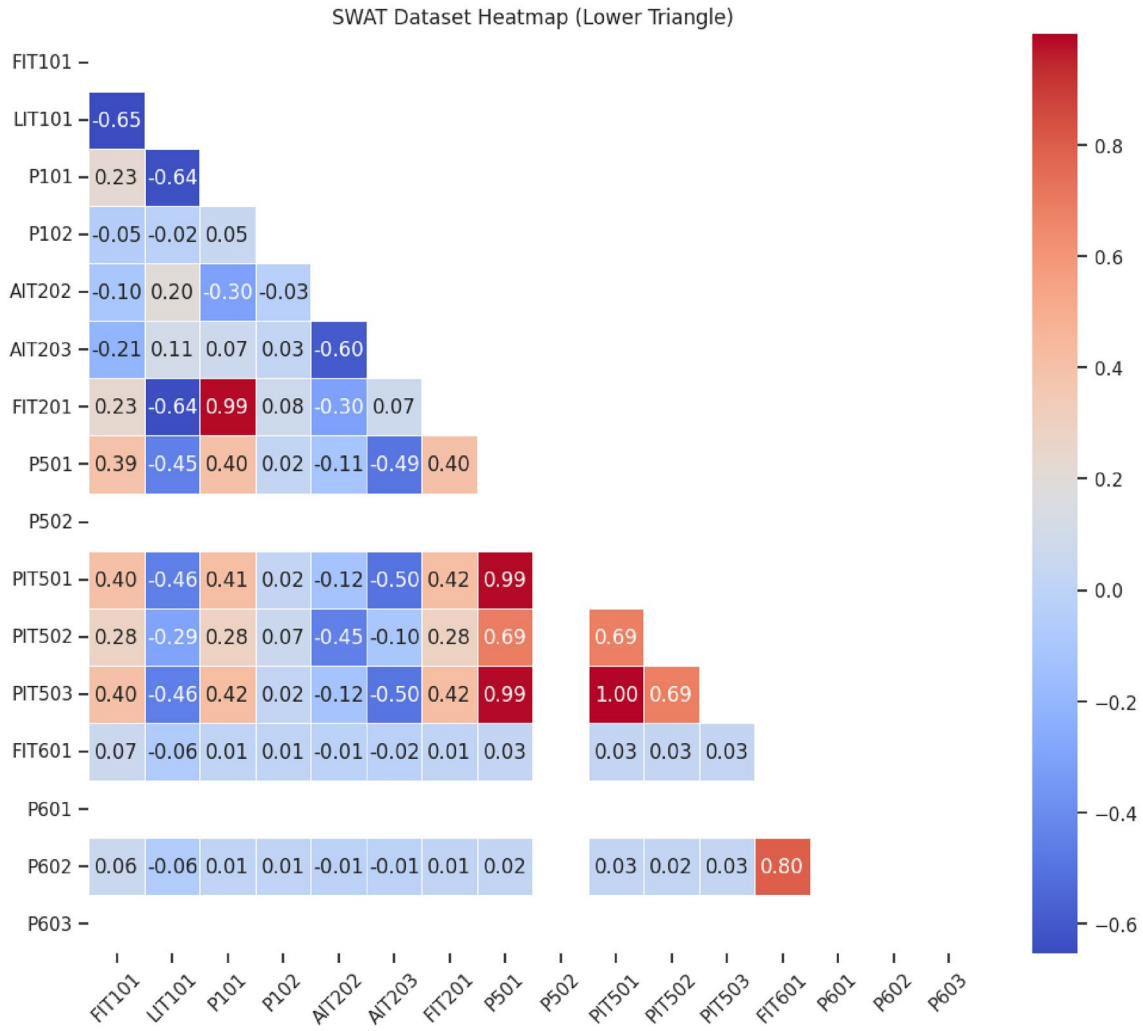


Fig. 3. Graphical representation of SWaT dataset.

moderate correlation was found between the characteristics and the labels; therefore, we retained all the features. After conducting variance research for the second task, it was found that features with tighter variances typically had a smaller range of possible values. Primarily, these features pertain to classifications related to motorized pumps and valves. Next, we removed those elements. We employed the K-S test in the third task to determine if continuous characteristics from the training and test datasets originated from the same statistical distribution. There were greater K-S statistics in the features of the chemical process sensor. Features with a K-S statistic score greater than 0.25 were therefore eliminated.

Feature extraction employs expert knowledge to extract higher-order features from the original features. Since ICSs are designed to manage industrial and essential processes, which often involve repetitive operations, the following outlines the tasks required to achieve those higher-order features. For instance, these restrictions must be considered by an ICS that monitors and regulates a water tank, modifying the input and output water valves in response to variations in the water tank's level. We recommend using autocorrelation and the GWO to identify these recurrent activities and extract features. In this work, we used correlation over a time window of L = 60 for the SWaT dataset and an L = 50 time window for the WADI dataset. By considering a data window rather than just a single data point, the model can identify temporal relationships and patterns that may indicate abnormalities. The fundamental idea of autocorrelation is represented by equation 18. The dense layer captures the intricate linkages and patterns in the dataset. The implied neural network model's depth is indicated by its layer size 2, and unsupervised anomaly detection is accomplished using AE layers. We employed three AE layers in the proposed work to depict the depth of the presented model. Learning the intricate interactions and nonlinearity of a neural network can be facilitated by using the activation function. By compressing the output values between 0 and 1, sigmoid functions, which measure abnormality, can help identify anomalies. The number of neurons in each layer determines the model complexity. The number of neurons helps understand the complexity of the network. We used 128 for AE and 256 for dense layers in this work.

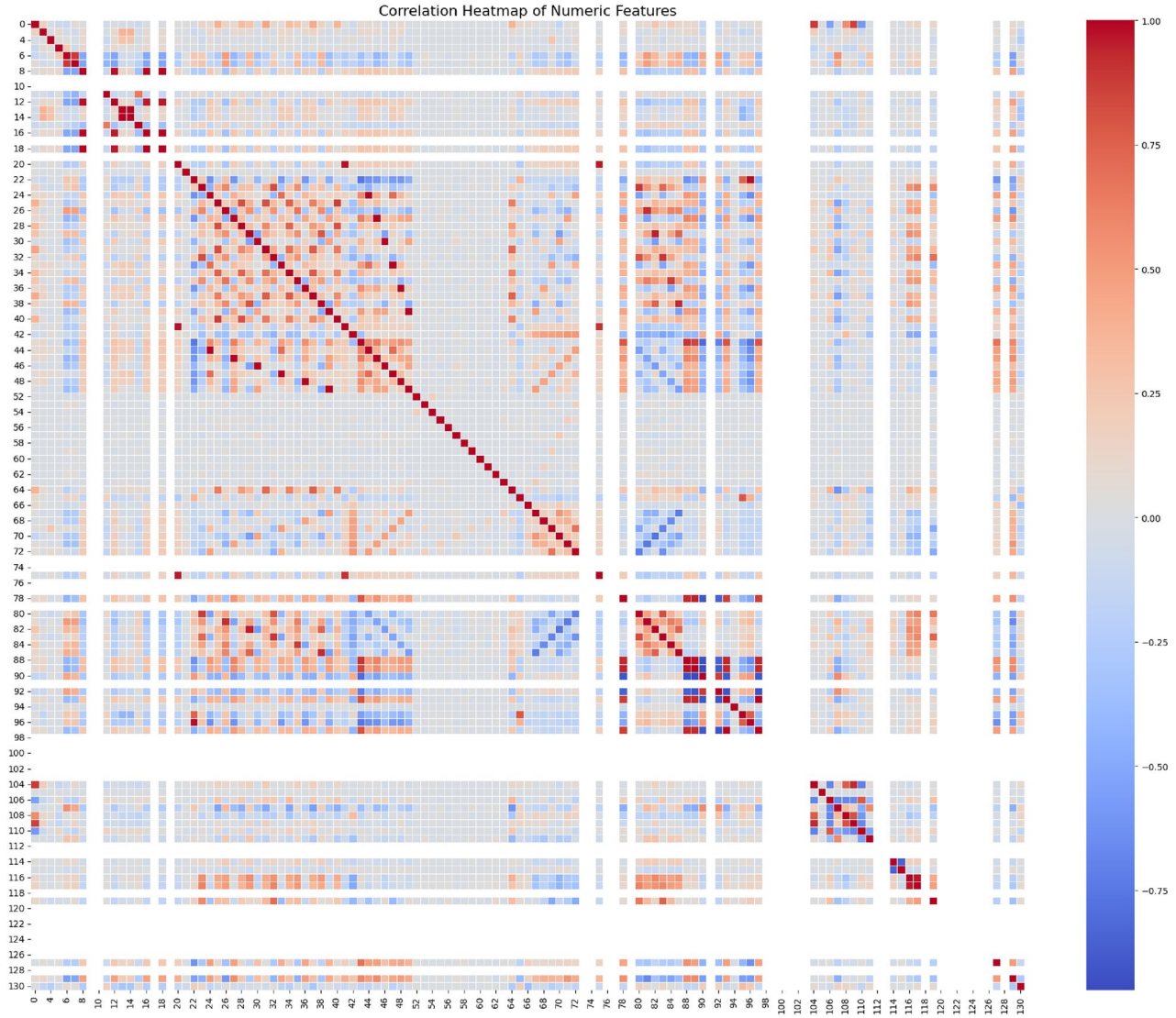


Fig. 4. Graphical representation of WADI dataset.

Training and tuning

To select, optimize, and train an appropriate model for anomaly detection, we looked at the procedures that need to be followed. Model selection is a crucial factor. The suggested model is the deep learning regressor model because it can identify complicated behaviors and performs better than NL approaches. Time series data is deemed appropriate aside from this. We recommend approaching the problem as a time-series data regression based on a statistical summary window, rather than using a classifier. This is because The first benefit of using a regression model is its ability to accurately simulate the temporal pattern that ICS sensor and actuator signals maintain. Second, a regression model with the right threshold can be used to get information about the sensor or actuator causing the anomaly, which helps to meet the desired accessibility criterion.

Training: We employed diverse strategies during this methodology's feature engineering and training phases to mitigate the effects of class imbalance. We specifically ensured the model had sufficient examples for learning by employing SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic samples of the minority class (attacks). We employed random undersampling of the majority class (normal data) to mitigate the bias towards normal data. We incorporated class weighting into the AE loss function and the GWO method during training, which imposes greater penalties on the model for misclassifying attack data. This enhances the model's sensitivity to rare assault events and reduces bias towards conventional data, augmenting its ability to detect anomalies. Furthermore, particularly in the AE model, anomaly detection techniques focus on reconstructing standard data patterns and identifying abnormalities through deviation. The selection of suitable features and the reduction of noise from the dominant class enhance the optimization of features using GWO, facilitating the detection of subtle anomalies.

Hyperparameter tuning process: The next step is to specify each hyperparameter's range of values so that training can be used to fine-tune them. Two groupings of these hyperparameters can be identified. The first

category of hyperparameters is those associated with the input data, while the second group consists of those particular to the DL model. To aid in choosing appropriate value ranges for the hyperparameters associated with the input data, we propose the following guidelines: To record behavior, the n window should be larger; h stands for future prediction; and m signifies the anticipated time step needed for that time step.

The next step is to specify the range of values for each hyperparameter, allowing training to fine-tune them. Two groupings of these hyperparameters can be identified. The first category of hyperparameters is those associated with the input data, while the second group consists of those particular to the DL model. To aid in choosing appropriate value ranges for the hyperparameters associated with the input data, we propose the following guidelines: To record behavior, the n window should be larger; h stands for future prediction; and m signifies the anticipated time step needed for that time step. The validation dataset is used to evaluate the performance of the hyperparameters, whereas the training dataset is utilized for the training process. More specifically, the hyperparameter quality is assessed using the validation reconstruction error, and the configuration that has the lowest error is the one that is optimized.

Monitoring the training reconstruction error to spot potential underfitting or overfitting issues in this activity is crucial. A significant reconstruction error during training and validation indicates underfitting or the deep learning algorithm's inability to accurately model the data. Another approach to mitigating overfitting is to compare the training reconstruction error to the validation reconstruction error; the former is significantly smaller, indicating that the model overemphasizes memorizing the training data over generalization. Some training datasets are used as the validation dataset in this approach. Consequently, during the training phase, it is unavoidable that some crucial information may be lost. When the hyperparameter search has identified the optimal hyperparameter values, the model should be trained using the combined training and validation datasets as the new training dataset. We can incorporate the previously missing data from the validation set to maximize the available data and improve the model's performance.

Data augmentation strategies can enhance the model's robustness to unforeseen anomalies. Generating synthetic data to enhance the training dataset is a form of data augmentation, which can improve the model's ability to generalize to previously unrecognized aberrant patterns. Noise injection and time-series transformation techniques, such as scaling, jittering, or shifting, can generate variations of existing data, enhancing the model's robustness against diverse anomalies. Conversely, SMOTE (Synthetic Minority Over-sampling Technique) and analogous methods can generate synthetic instances of infrequent attack events. Adversarial training techniques, which involve minor perturbations of input data, can help the model recognize novel, previously unobserved assault patterns. Future endeavors will incorporate analogous data augmentation methodologies to enhance the model's robustness in dynamic, real-world environments where unforeseen anomalies may arise.

Unveiling anomalies

Algorithm 2 utilizes AE for Unveiling anomalies and demonstrates the hidden layer property of AE. It shows that the studied dropout methodology is predicated on uniform distribution. It is essentially a continuation of the traditional dropout technique to improve the weights of the pertinent feature maps. The first step is the error prediction between ground truth and neural network, and mathematically is represented in equation 9, while equation 8 is used to find the MSE. AE is a DL-based unsupervised approach that yields outcomes close to the input and is used for feature extraction from a larger amount of data. There is a compromise between an encoder and a decoder in an AE approach. Here, an encoder reduces the dimension from the input to the latent space z. At the same time, the decoder is useful in generating the output, and the output is similar to the input, which demonstrates the effectiveness of feature extraction. After successful feature extraction, a noticeable difference appears between the input and the output.

The encoder and decoder's edge and bias are computed and optimized to the training dataset during the training phase. As a result, the AE cannot properly rebuild an input if it differs from the training input. The reconstruction error, also known as RE, is higher in this instance than in a typical scenario. This feature is useful for identifying anomalies. Protocol header features have been employed in traditional studies to examine anomaly detection. The ICS network traffic, however, remains largely unchanged. Thus, the straightforward whitelist rule adequately represents the typical behavior of the packet header data. From this perspective, header anomaly detection can be performed without the need for deep learning. The accuracy attained when utilizing the whitelist rule is higher than when a deep learning model predicts a field value that is always similar due to system design. Therefore, failing to consider the payload abnormalities reduces the effectiveness of using deep learning for anomaly identification.

Suppose the model input is $x_{t:t+T}$ with T length. For the model output, y_t and z_t were used for the reconstructed and predicted time series, respectively. The mean square error (MSE) was used to calculate the difference between the input and output. Mathematically, it is given as

$$\mathrm{MSE} = |x_t^m - \hat{x}_t^m|^2 \tag{11}$$

In this Eq. 11, x_t is the input series, x_mth is the dimension value, and $\hat{x}_t^m$ is the reconstruction output. Now in $L_s = L_c + L_p$ L is the loss function used for calculating MSE, L_c is the reconstruction part loss L_p is the prediction part loss.

The error from the two sides is separately calculated and summed to obtain E_t, which represents the overall error. The following steps are to calculate

$$e_t^{cons} = |x_t - y_t| \tag{12}$$

$$e_t^{pred} = |x_t - z_t| \tag{13}$$

$$E_t = e_t^{cons} + e_t^{pred} \tag{14}$$

An approximately moving average method has been used to overcome the sudden error.

$$SE_t = \alpha E_t + (1-\alpha) SE_{t-1} \tag{15}$$

where $SE_0 = 0$; and $\alpha = 1 - exp\frac{ln0.5}{H}$, to decrease the false positive, used power technique

$$PE_t = \frac{1}{M}\sum_{i=1}^{m} |SE_t^i|^p \tag{16}$$

In the case of PE_t being more than V_{thre}, the *t* time observation is justified as an anomaly. Here, V_{thre} is chosen from PE_t. For this procedure, the algorithm steps are given in...

$$A_t = \begin{cases} 1, & \text{if } PE_t \geq V_{\text{thre}} \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

$$p_m = 0.5e^{-\frac{10t}{T}} + 0.01 \tag{18}$$

$$\mathcal{Z} = \max\left\{1, \left[D - \left(\frac{t}{T}\right)^{\mathcal{A}} \times p_m\right]\right\} \tag{19}$$

In this equ 19, $\mathcal{Z}$ represents wolves pack, $\times p_m$ is a nonlinear function which is given as $p_m = 0.5e^{-\frac{10t}{T}} + 0.01$; *t* is current, and *T* is maximum iteration/s. Mathematical proof of the dropout layer is given as

$$\begin{aligned} a^{(l)} &= \text{Ber}(p) \\ (u^{(l)})'' &= a^{(l)} \times u^{(l)} \\ v_i^{(l+1)} &= \sigma_i^{(l+1)}(u^{(l)})'' + c_i^{(l+1)} \\ u_i^{(l+1)} &= f(v_i^{(l+1)}) \end{aligned}$$

In this equation, *l* is for the hidden layer, and *i* is for hidden units. Here, *l* is linked with $v_i^{(l)}$ and $u_i^{(l)}$; those are input and output vectors, respectively. *Ber* is short term for Bernoulli. Weights $\sigma_i^{(l)}$ and biases $c_i^{(l)}$ are linked with *l*. In this approach, a random number vector $a^{(l)}$ primarily yielded from the *Ber* distribution and is then multiplied with $(u^{(l)}$ to generate $(u^{(l)})''$ which is considered as thinned output. In the next step, the output of this thin output is used as input in the next layer to compute $v_i^{(l+1)}$. Using the *f* activator function, the process is repeated for each layer. Notably, the Bernoulli distribution yields an output of either "0" or "1," indicating that a certain hidden unit is either fully eliminated or carried forward.

Require: Hidden layer activations u^l, layer indices $\lambda = \{l_1, ..., l_m\}$
Ensure: Updated layer outputs with dropout
1: **for** each $l_i \in \lambda$ **do**
2: Sample $a^l \in [0, 1]$ from uniform distribution
3: Apply dropout: $\tilde{u}^l = a^l \cdot u^l$
4: Compute: $v_i^{l+1} = \sigma_i^{l+1} y^l + c_i^{l+1}$
5: Update: $u_i^{l+1} = f(v_i^{l+1})$
6: **end for**

Algorithm 3. Improved AE with dropout layer for anomaly detection.

Require: Dataset T, pop, F, pos, x_t, y_t, Z_t, V_{thre}
Ensure: Highly correlated features and anomaly labels A_t
1: **Step 1:** Invoke **Algorithm 1** with T, pop, F, pos to obtain selected features
2: **Step 2:** Apply **Algorithm 2** or optionally **Algorithm 3** for anomaly detection
3: **return** Selected features and detected anomalies A_t

Algorithm 4. Proposed framework: GWO + AE-based anomaly detection.

In this Algorithm 3, at the initial stage, the λ begins $\forall$ hidden layers with weights ≥ 0.5. Secondly, a^l vector begins with uniform distribution and random numbers. The thinned outputs are then calculated by pairwise multiplication of a^l and u^l, followed by v_i^{l+1}. Lastly, the next layer output like $(l+1)$, are guessed and this process goes on repetition $\forall$ layers.

When attackers aim to harm an ICS or interfere with its regular operations, they typically damage sensors or actuators. The next step is to identify the variables (ICS devices) under assault after using the technique described in the previous section to detect anomalies when attacks occur correctly. This enables the staff to identify anomalies and take action to restore the system to regular operation. Put differently, the suggested dropout technique modifies the weights of some of the units (with weights> 0.5) in addition to dropping some of the units and connections from the network like traditional dropout. This task optimizes the weights of the network's most critical features while preventing overfitting. Consequently, this improves the architecture's precision and speeds up its convergence.

In this work, GWO is useful for feature engineering because it assists in formulating the multi-objective feature selection challenge. Given a set of feasible non-dominated solutions, the optimal solution for each wolf will be found. The fitness function of the participating wolf swarm is expressed mathematically below in this context. The detailed algorithm is presented in Algorithm 1, and the proposed algorithm is explained in Algorithm 4.

Reducing feature numbers and minimizing the classification error rate are the key challenges in feature selection in the given SWaT dataset. Making the best choices is challenging when trade-offs exist between two or more competing goals. A single objective problem with numerous constraints may not adequately represent this topic. In this situation, minimizing or maximizing the set of objective functions requires multi-objective optimization, which functions under a specific set of constraints. This approach is helpful for the subset of features and minimizes the classifier error rate ϵ. Previously, researchers have used various methods, including Hamming loss[63], accuracy[64], and ranking loss[65], to evaluate classifier performance. This paper uses accuracy as an evaluation criterion to determine the classification error rate of the grey wolf. To reduce the ϵ fitness function, it is given below in Eq. (20).

$$\epsilon = \frac{(FP + FN)}{TP + TN + FP + FN} \tag{20}$$

This function determines which feature set minimizes the classification error in feature selection by testing every feasible subset of features during the evolutionary training phase. *FP*, *FN*, *TP*, *andTN* will be discussed later.

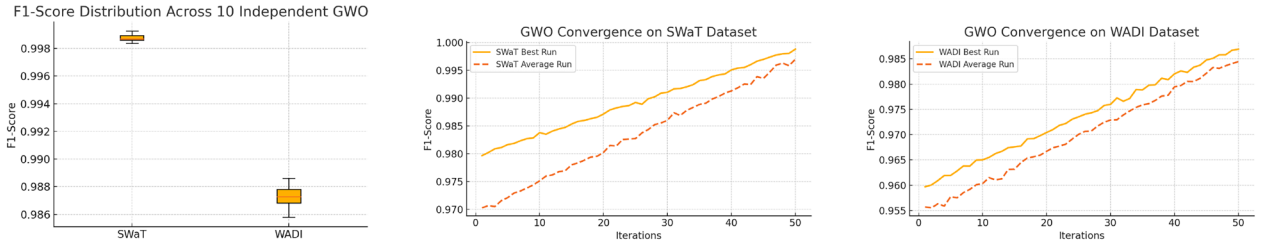
Next is to take the multi-objective fitness function F in which F_1 reduces the ϵ while F_2 reduces the features[66].

$$F = \begin{cases} EMPTY\epsilon \cdot F_1 \\ EMPTY\gamma \times \frac{F'}{F} + (1-\gamma) \times \frac{\epsilon_c^{F'}}{\epsilon_c^{F}} \quad (F_2)EMPTY \end{cases} \tag{21}$$

In this, γ is a constant value between [0,1], F' is the number of features, and F is the total number. Equation 21 reduces the classification error and selects highly correlated features in the dataset. It balances the highly correlated features. The classification error rate during feature selection is represented by $\epsilon_c^{F'}$, and the error rate is defined by ϵ_c^{F}. This error rate is involved in all classified features.

Simulation and results

This section highlights the proposed architecture's simulation and outputs. The case was evaluated using the SWaT and WADI datasets. Precision, recall, accuracy, and F1-score attacks were used for model detection. We also noticed a false positive rate in this experiment. The detected attack shows the number of attack detectors using the presented approach, and recall, F1-score, and precision were used for secondary metrics. For a secure ICS environment, high recall and low FPR are essential. For clarification, the results are compared with approaches such as AE and GWO, as well as the collaboration between AE and GWO.



(a) F1-score distribution for SWaT and WADI datasets (b) GWO convergence on SWaT dataset (c) GWO convergence on WADI dataset

Fig. 5. Performance and convergence analysis of the GWO across SWaT and WADI datasets.

Method	Prec.	Rec.	F1
LSTM[47]	0.961	0.90	0.93
SVM, ID-CNN[48]	0.932/0.97	0.70/0.79	0.80/0.87
DT, RF, ANN[49]	0.994/0.995/0.995	0.979/0.982/0.992	0.991/0.995/0.980
Multi-Model[37]	0.996	0.996	0.996
Deep FL[39]	0.951	0.953	0.951
LSTM[50]	0.995	0.994	0.982
DT[51]	0.974	0.935	0.835
AFAD[52]	0.920	0.91	0.91
DAICS[36]	0.918	0.862	0.889
STADN[53]	0.999	70.79	0.83
GDN[38]	0.993	68.12	0.81
1D CNN+AE[54]	0.916	0.731	0.754
MAD-GAN[55]	70.00	95.40	0.81
UAE freq.[54]	0.911	0.860	0.885
LSTM[56]	N/A	N/A	0.958
GNN[57]	98.29	70.84	0.82
GTN[58]	98.19	70.40	0.82
MTS-DVGAN[40]	99.00	66.93	79.87
ESTAD[57]	99.25	76.01	0.86
Proposed	**0.9996**	**0.9989**	**0.9987**

Table 3. SWAT dataset performance comparison. Significant values are in bold.

Dataset	Approach	Precision	Recall	F1-Score
WADI	AE[59]	N/A	54.2	N/A
	DAICS[60]	0.9083	0.7205	0.8036
	CNN-1D[61]	0.7935	0.4123	0.54
	USAD[62]	0.84	0.80	0.79
	STADN[53]	98.49	45.57	0.62
	GDN[38]	97.50	40.19	0.57
	LSTM-VAE[56]	N/A	N/A	0.459
	GNN[57]	93.42	43.23	0.59
	GTN[58]	87.39	52.38	0.65
	MTS-DVGAN[40]	96.23	99.51	97.84
	MAD-GAN[55]	53.75	74.92	0.62
	ESTAD[57]	74.01	49.22	0.59
	Proposed	**0.9939**	**0.9913**	**0.9873**

Table 4. WADI dataset performance comparison. Significant values are in bold.

Figure 5 illustrates the efficiency of the GWO on the SWaT and WADI datasets. Figure 5a illustrates the distribution of F1-scores over 10 independent trials. This indicates that the optimisation process is stable and reliable. Figure 5b, c demonstrate the convergence of the GWO across 50 iterations for the SWaT and WADI datasets, respectively. The charts indicate that the optimiser can progressively enhance detection performance until it identifies solutions with elevated accuracy and minimal variation.

We have incorporated a comparison of GWO-based feature selection with traditional optimization techniques, such as grid search and random search, alongside machine learning models, including Decision Trees and Support Vector Machines (SVMs), to provide context for findings. Conventional approaches often exhibit disadvantages, including slow convergence, limited applicability to novel data, and susceptibility to overfitting. Here, findings indicate that the GWO-based feature selection method significantly surpasses conventional methods in terms of accuracy and computational efficiency. Conventional methods, such as grid and random search, require comprehensive evaluations of hyperparameters, which can be computationally expensive. In contrast, GWO attains superior performance with less testing by adapting its search strategy and efficiently diminishing the feature space.

GWO was helpful in feature engineering and selecting optimal or highly correlated features from the dataset. It assists in enhancing the feature set selection and reducing the error rate. A total of 51 features were provided in the SWaT dataset, of which GWO assisted in selecting highly correlated features. It selected the useful features between sensors and actuators like connection duration, protocol type, number of unsuccessful attempts, etc.

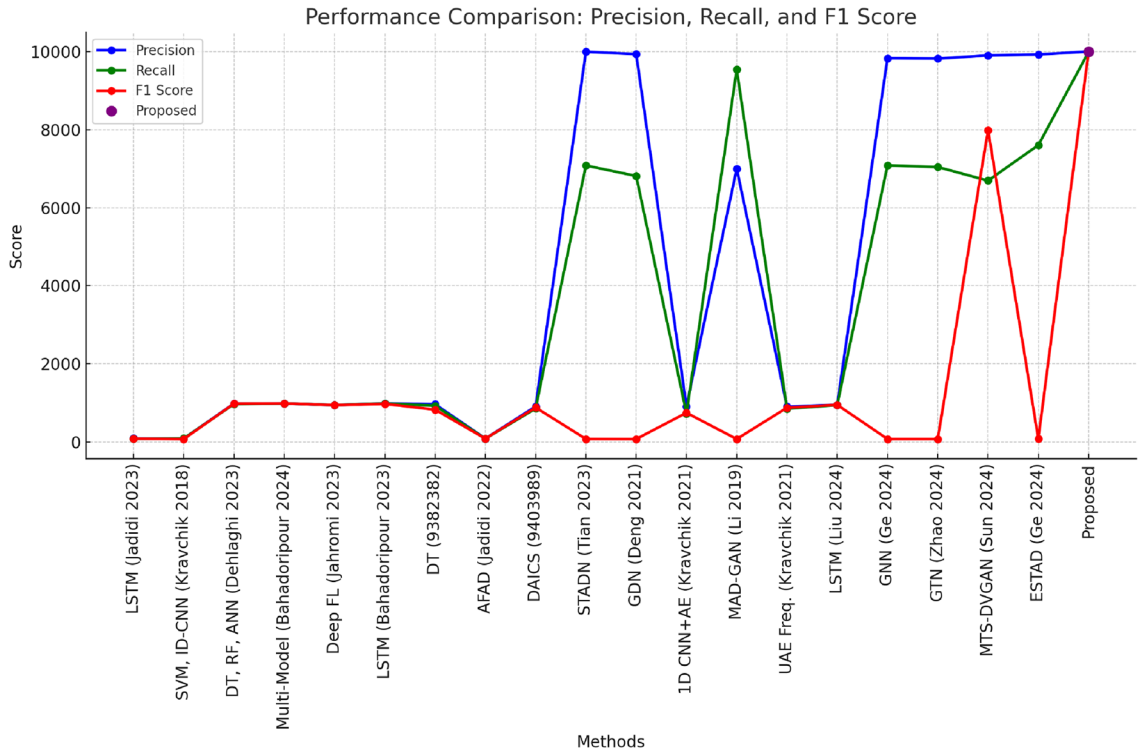
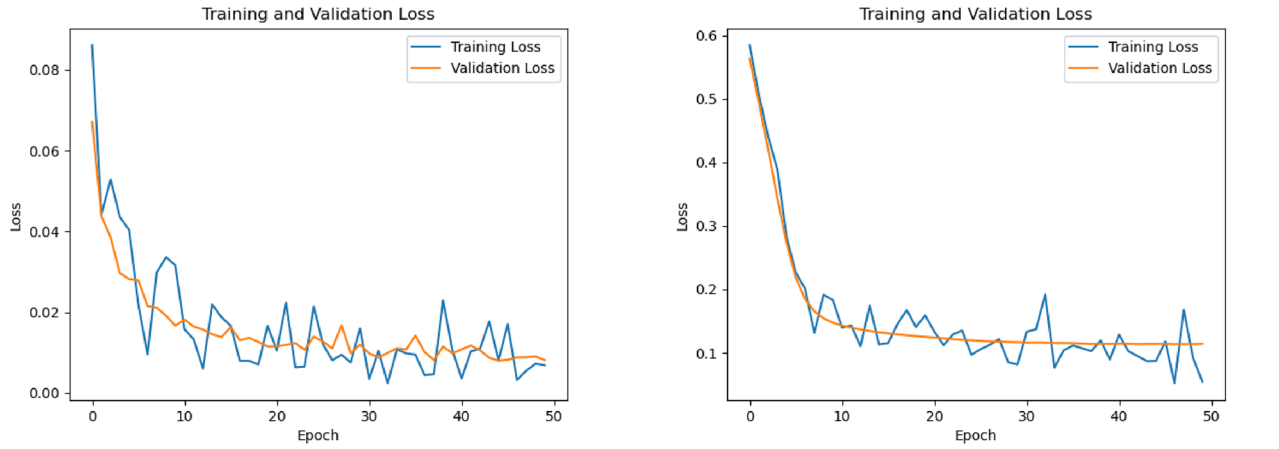


Fig. 6. Graphical representation of SWaT dataset.



(a) SWaT: Training and validation loss (b) WADI: Training and validation loss

Fig. 7. Training and validation loss curves for the SWaT and WADI datasets.

The performance of the proposed collaborative approach for anomaly detection in the SWaT dataset is studied. In terms of outcomes, we achieved a precision of 0.9996, a recall of 0.9989, an F1-score of 0.9987, and an accuracy of 0.9996. The proposed collaborative model (AE+GWO) improved the outcomes. We achieved the highest accuracy, precision, recall, and F1-score, accompanied by a very low false positive rate of 0.0001. Similarly, the WADI dataset contains noise and NaN values during evaluation. The first row in the dataset created an issue of float-to-string conversion, which was not necessary for the evaluation process, so we deleted that row without any important information. During the simulation process, an error occurred with nan values in the dataset. However, upon rechecking, no nan values were found in the dataset. We resolved this issue by using the "handle missing values data = data.fillna(0)" in the proposed algorithm. Now, it was ready to simulate and yielded sufficient and better precision results, with a recall of 0.9913, F1-score of 0.9873, accuracy of 0.9974, and a very low FPR of 0.0002. Further, we have compared the SWaT and WADI outcomes with the work state in

Tables 3 and 4 respectively. Figure 7 represents the training and validation loss of the SWaT and WADI datasets, respectively. The comparison results are presented in Tables 3 and 4.

In the Fig. 6, we present a detailed comparison of the performance of various anomaly detection methods on the SWaT dataset, focusing on three key metrics: Precision, Recall, and F1 Score. The results, as shown in Fig. 6, highlight the effectiveness of the proposed method relative to other state-of-the-art approaches. The Proposed method achieves the highest performance in all three metrics, with a Precision of 0.9996, a Recall of 0.9989, and an F1 Score of 0.9987, demonstrating its superior ability to identify anomalies while accurately minimizing false positives and false negatives.

Methods such as LSTM, SVM with ID-CNN, and Multi-Model also show strong performance, with high Precision and Recall values, although they do not match the overall effectiveness of the proposed approach. Other methods, such as STADN, GDN, and 1D CNN+AE, exhibit lower performance, particularly in terms of Recall and F1 Score, indicating limitations in handling complex anomaly detection tasks in industrial control systems. This comparison underscores the proposed method's robustness and reliability, highlighting its ability to achieve high performance across various evaluation metrics consistently. The superior results also reflect the optimization approach incorporated into the model, contributing to its effectiveness in real-world anomaly detection scenarios.

Conclusion and future work

This work presented a collaborative approach for anomaly detection in ICS. Specifically for feature extraction and anomaly detection in real-time communication, the model utilizes deep learning and multi-objective optimization techniques. The GWO approach, which uses deep learning for anomaly detection in ICS, is novel. It extended anomaly detection compared to state-of-the-art works in ICS. This approach has been evaluated using metrics such as precision, recall, and F1-score, which are commonly employed in the literature review for classification purposes. However, other metrics, such as FPR and accuracy, have been evaluated in this work, which is uncommon in practice and demonstrates the model's effectiveness in similar perspectives, including precision, recall, and F1-score. Model evaluation on three benchmark ICS datasets, SWaT and WADI, reveals high precision, recall, and F1-score, along with a low positive rate. It demonstrates the effectiveness of the model. The proposed approach possesses numerous challenges, despite its efficiency. Initially, scalability remains an issue, as employing the model in extensive ICS configurations with additional sensors and increased data may necessitate greater computational resources. Secondly, the methodology requires labeled attack data for training, which is difficult to obtain in reality due to the infrequency of cyberattacks and the challenge of acquiring accurate labels. Finally, there is a potential for overfitting to the SWaT and WADI datasets, which may diminish their applicability to other ICS systems that exhibit distinct behaviors or attack patterns. The current approach prioritizes the identification of single-stage anomalies while disregarding multi-stage or coordinated intrusions that necessitate more advanced techniques. We propose including temporal or sequential analysis to elucidate the interdependencies across different phases of an assault, hence enhancing the model for multi-stage attacks. RNNs and LSTM networks are two methodologies employed to predict the progression of attacks over time. Incorporating graph-based anomaly detection methods or behavioral profiling aids in identifying coordinated patterns throughout various stages of an attack. A subsequent study will examine these strategies to enhance the detection capabilities of the proposed model for multi-stage and coordinated cyberattacks. Future studies will explore adaptive hyperparameter adjustment techniques to enhance model efficiency and accuracy. Contemporary hyperparameter tuning methods, such as grid search and random search, are typically static and may not accommodate the dynamic nature of the data or the model's fluctuating performance throughout training. Bayesian optimization, GA, and reinforcement learning-based tuning exemplify adaptive approaches that can dynamically adjust hyperparameters based on the model's performance during different training phases. By continuously optimizing parameters such as learning rate, regularization strength, and optimization algorithm configurations, these methods can improve model convergence, reduce computational expenses, and yield more precise results. Subsequent iterations of the model will prioritize the incorporation of adaptive hyperparameter optimization.

Data availibility

The iTrust centre, Singapore University of Technology and Design, provided data upon the authors' request. As the data is confidential for the authors and authors are not allowed to share the data, it can be requested for research purposes from the following webpage https://itrust.sutd.edu.sg/research/publications/ by selecting iTrust Labs then datasets, and then the required dataset. Further queries can be contacted corresponding author.

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Author contributions

M.M.A Original writing, methodology, visualization; L.C.D.S. Supervision, editing and writing, conceptualization, R.A.A.H.M.A. Supervision, methodology, conceptualization, visualization and A.T. Supervision, editing and reviewing, methodology, Funding.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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